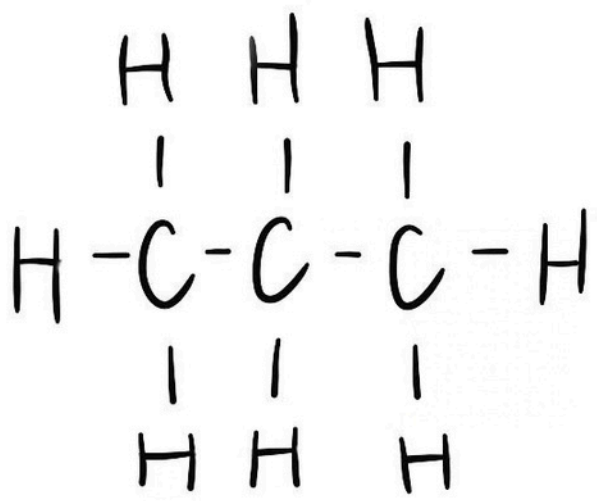
PROPANE

CARD 01 - C_3H_8

1. **MOLECULAR FORMULA**

C_3H_8

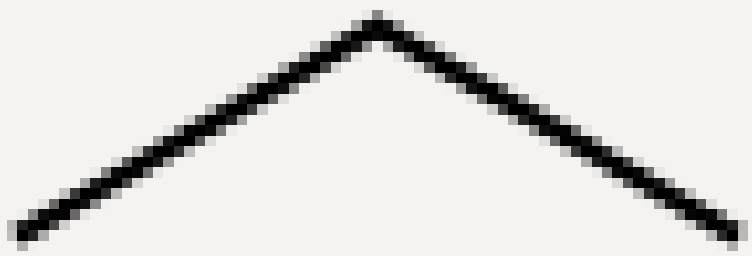
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$CH_3CH_2CH_3$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

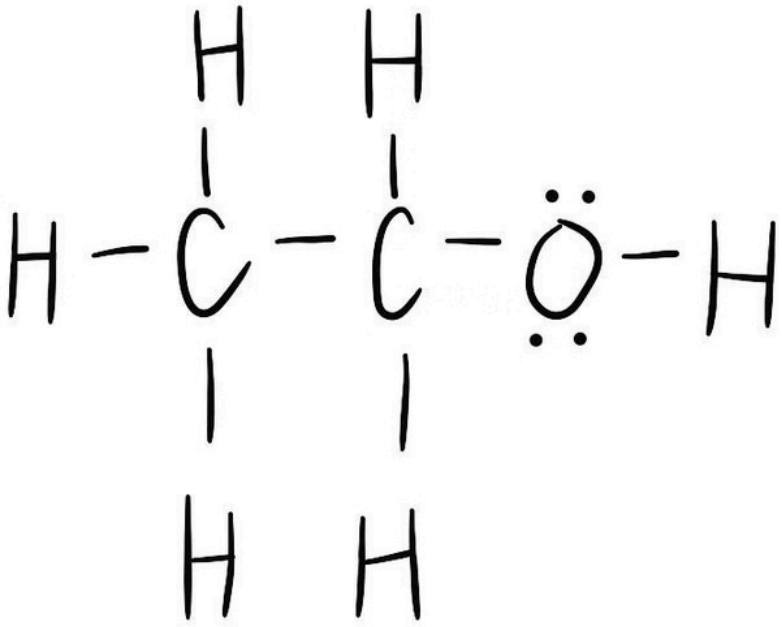
ETHANOL

CARD 02 - C_2H_6O

1. **MOLECULAR FORMULA**

C_2H_6O

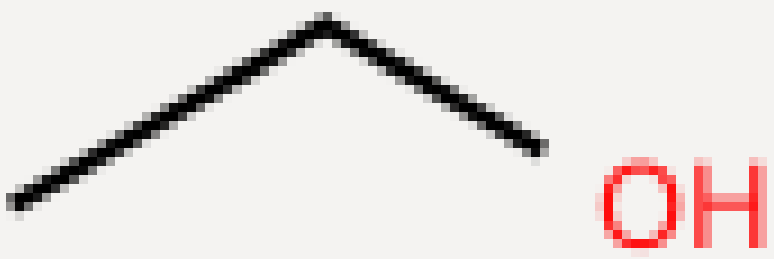
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

CH_3CH_2OH

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

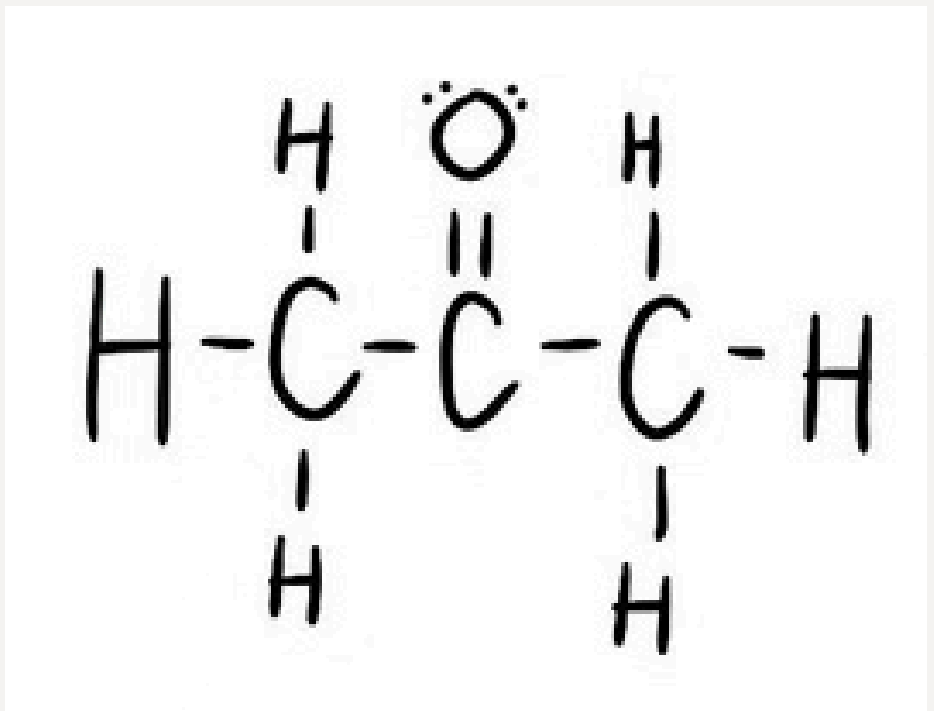
ACETONE

CARD 03 - C_3H_6O

1. **MOLECULAR FORMULA**

C_3H_6O

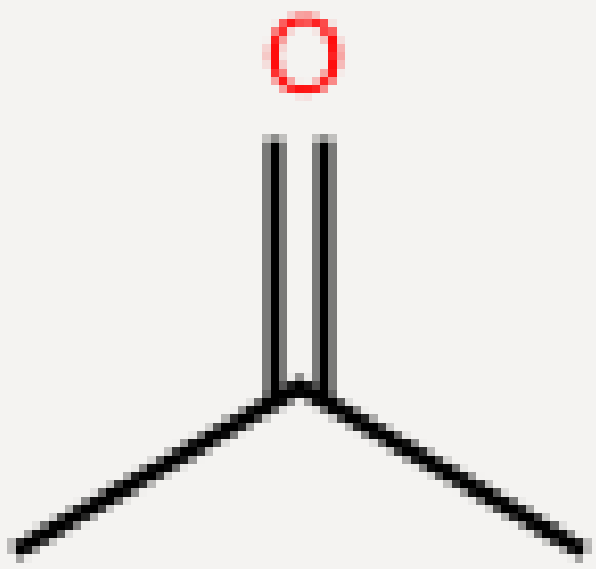
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

CH_3COCH_3

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

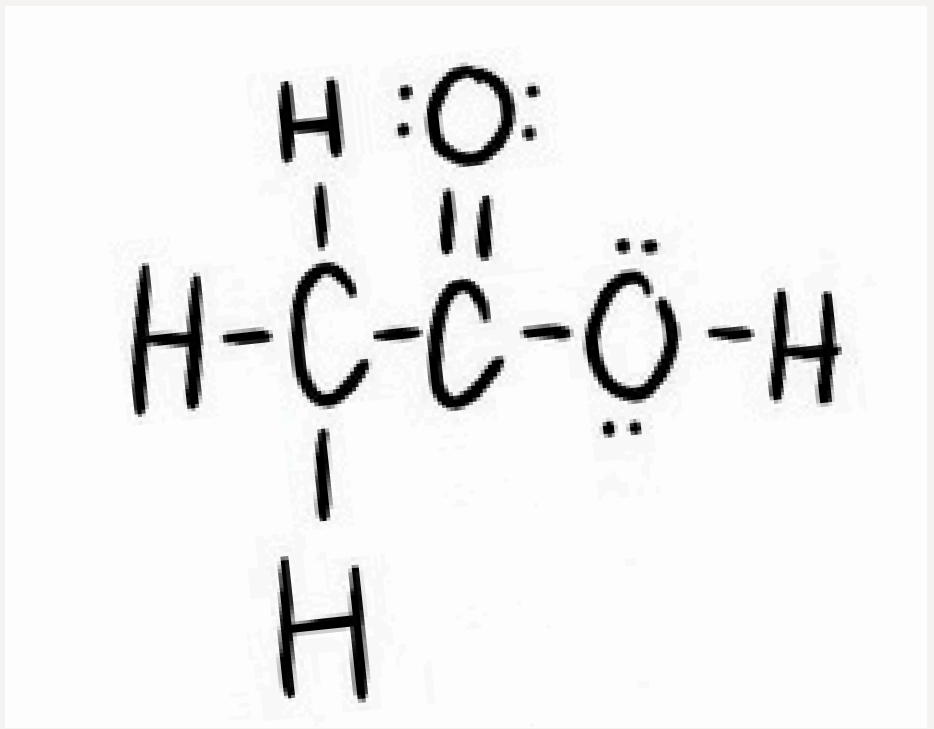
ACETIC ACID

CARD 04 - $C_2H_4O_2$

1. **MOLECULAR FORMULA**

$C_2H_4O_2$

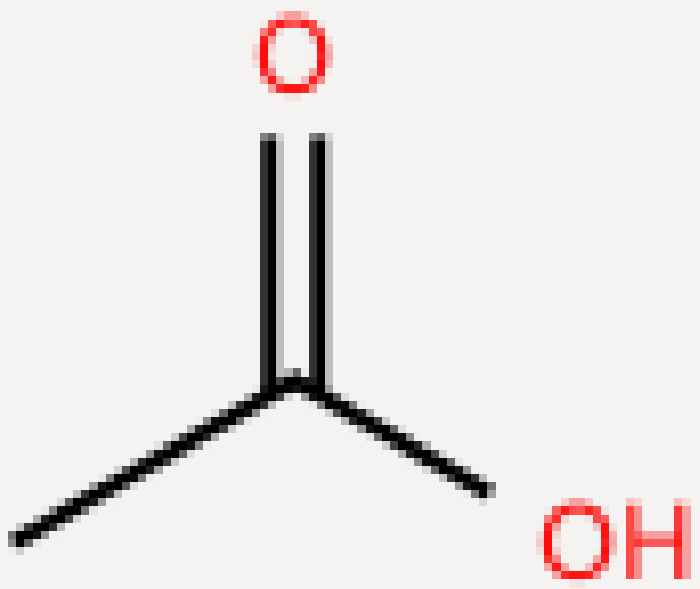
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

CH_3COOH

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

- [ ] ATOM COUNT VERIFIED
- [ ] VALENCE CHECKED
- [ ] BOND ODER CHECKED
- [ ] SAME CONNECTIVITY CONFIRMED

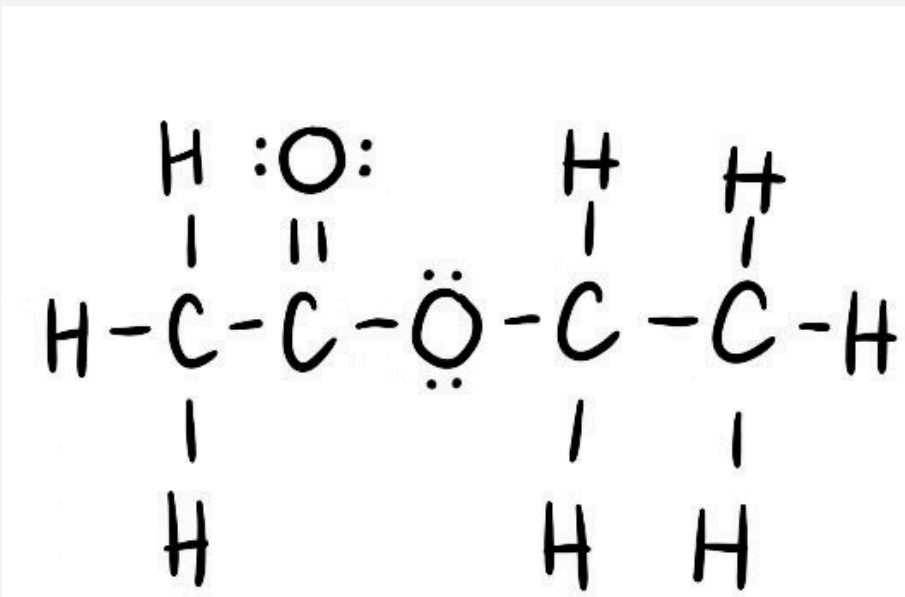
ETHYL ACETATE

CARD 05 - $C_4H_8O_2$

1. **MOLECULAR FORMULA**

$C_4H_8O_2$

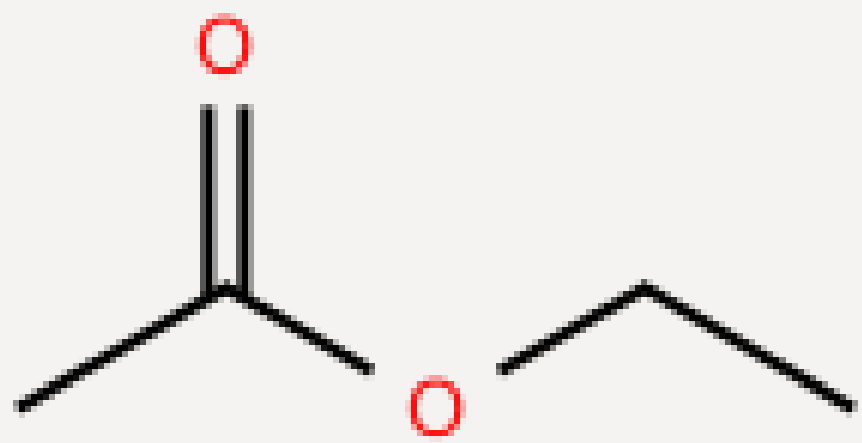
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$CH_3COOCH_2CH_3$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

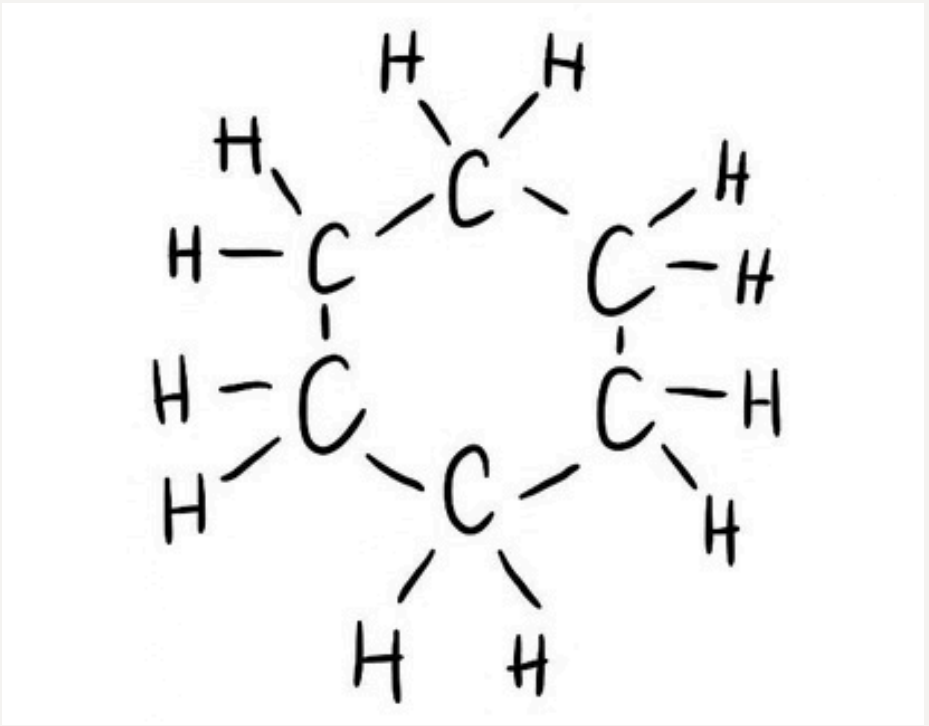
CYCLOHEXANE

CARD 06 - C_6H_{12}

1. **MOLECULAR FORMULA**

C_6H_{12}

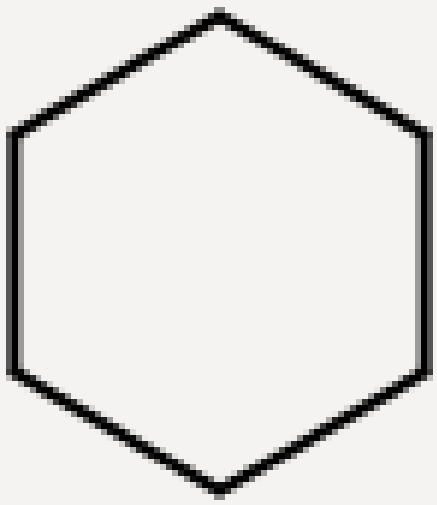
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$(CH_2)_6$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

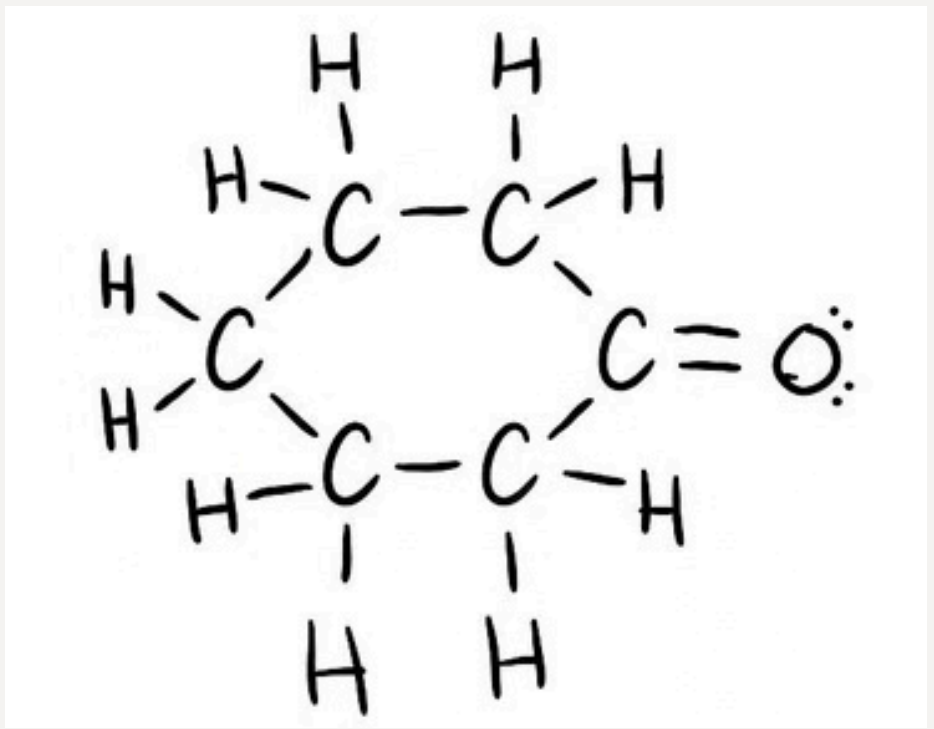
CYCLOHEXANONE

CARD 07 - $C_6H_{10}O$

1. **MOLECULAR FORMULA**

$C_6H_{10}O$

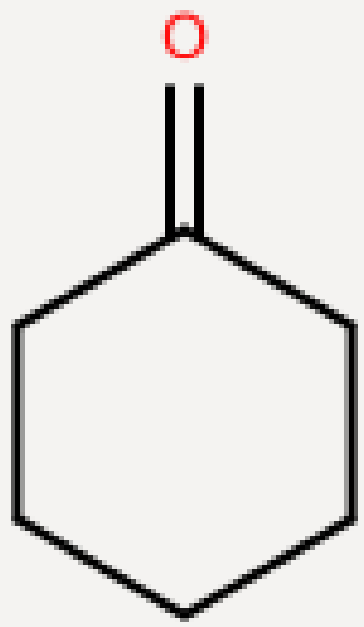
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$(CH_2)_5CO$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

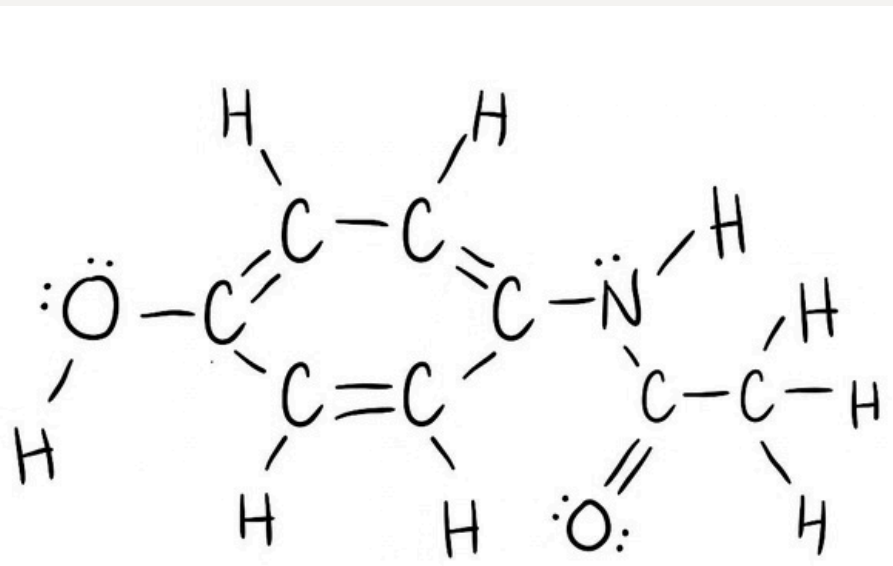
ACETAMINOPHEN

CARD 08 - $C_8H_9NO_2$

1. **MOLECULAR FORMULA**

$C_8H_9NO_2$

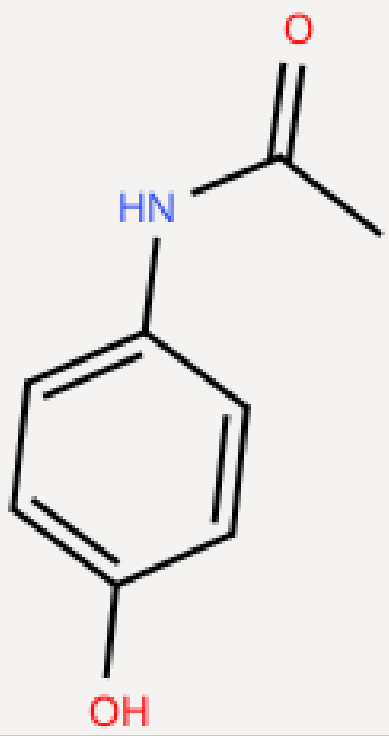
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$HOC_6H_4NHCOCH_3$

4. **BOND-LINE FORMULA**



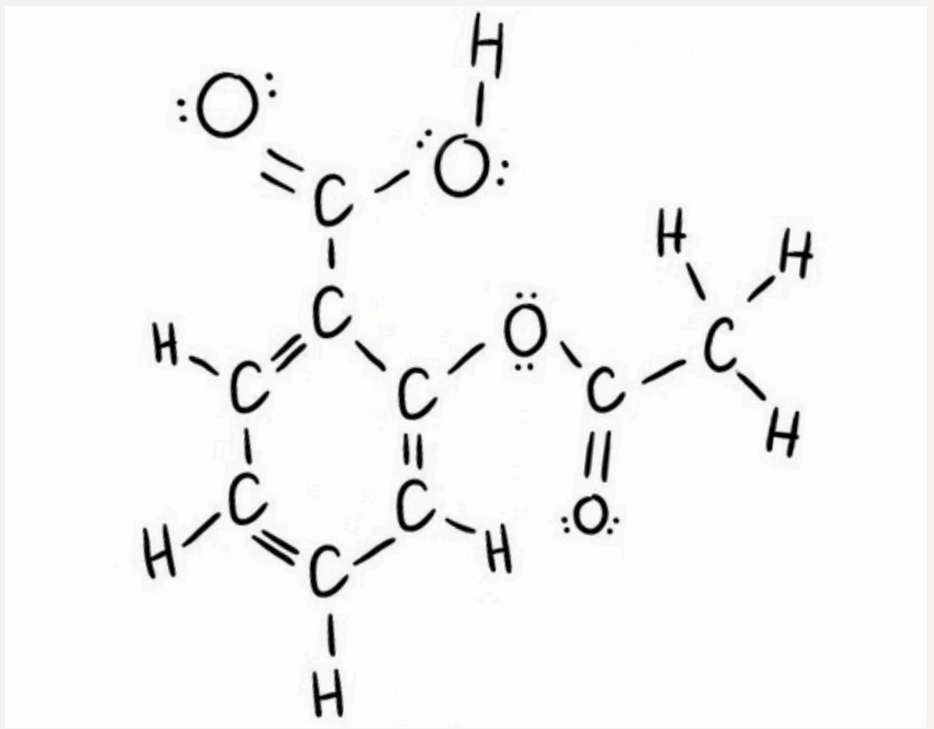
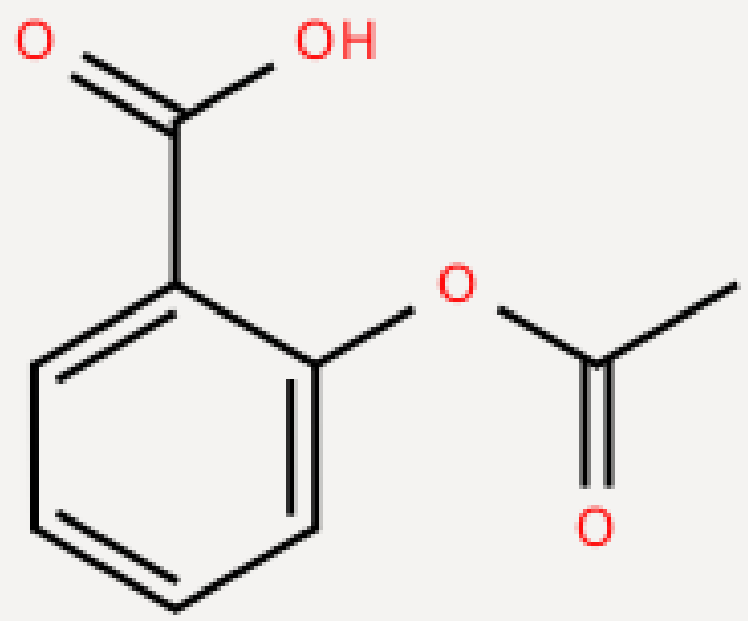
VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

ASPIRIN

CARD 09 - $C_9H_8O_4$

1. **MOLECULAR FORMULA**	2. **LEWIS STRUCTURE**
$C_9H_8O_4$	
3. **CONDENSED STRUCTURAL FORMULA**	4. **BOND-LINE FORMULA**
$CH_3COOC_6H_4COOH$	

VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

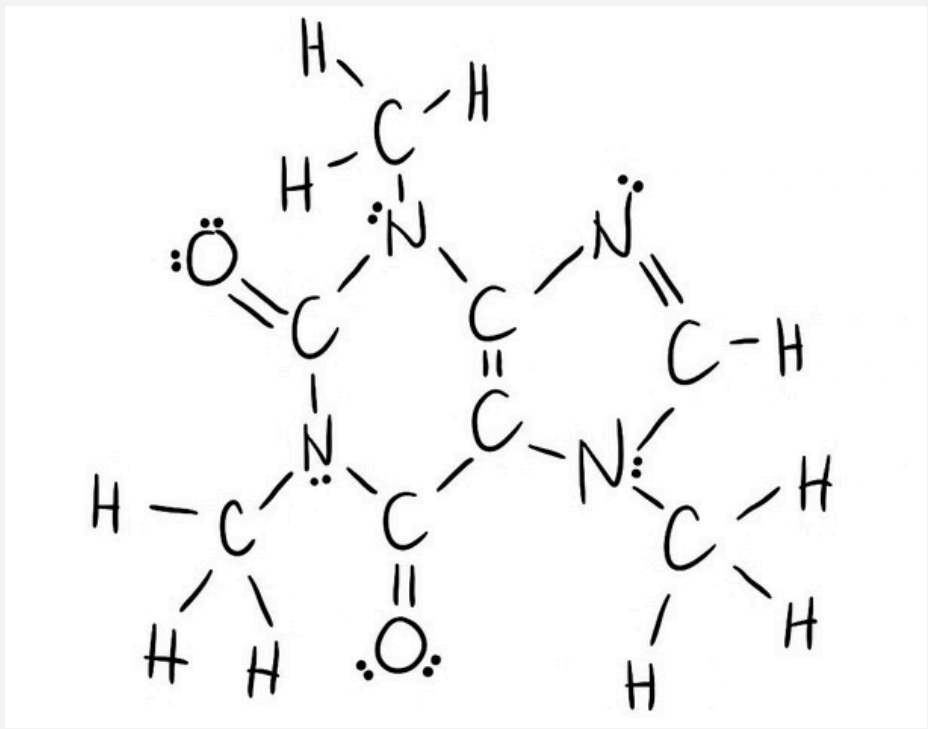
CAFFEINE

CARD 10 - $C_8H_{10}N_4O_2$

1. **MOLECULAR FORMULA**

$C_8H_{10}N_4O_2$

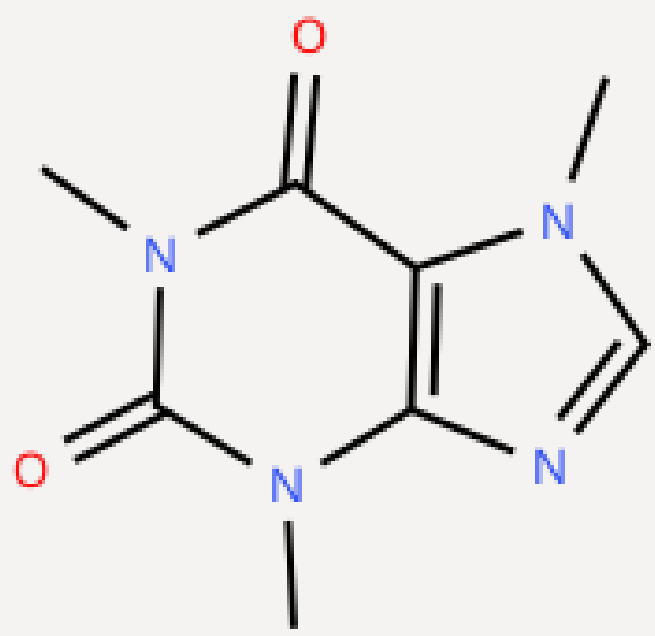
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$(CH_3N{-}C{=}O)_2C{=}C(N{=}CH{-}NCH_3)$

4. **BOND-LINE FORMULA**



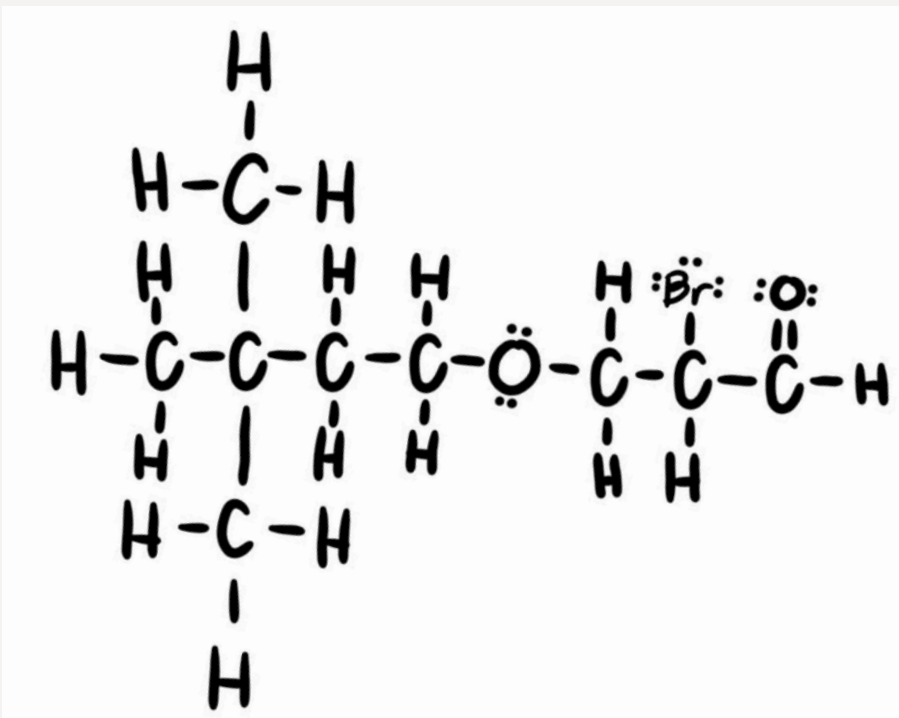
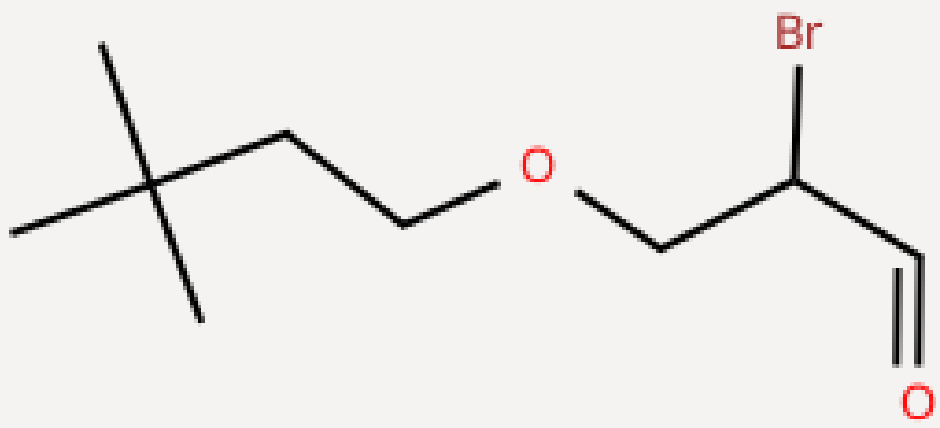
VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

2-BROMO-3-(3,3-DIMETHYLBUTOXY)PROPANAL

CARD 11 - PROBLEM 1

1. **MOLECULAR FORMULA**	2. **LEWIS STRUCTURE**
$C_9H_{17}BrO_2$	
3. **CONDENSED STRUCTURAL FORMULA**	4. **BOND-LINE FORMULA**
$(CH_3)_3CCH_2CH_2OCH_2CH(Br)CHO$	

VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

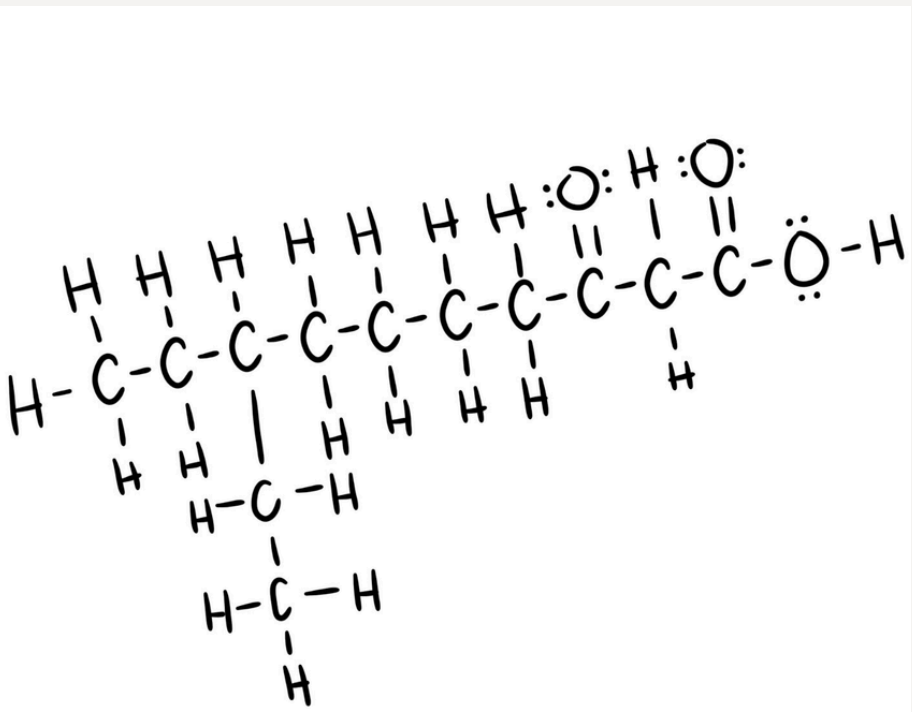
8-ETHYL-3-OXODECANOIC ACID

CARD 12 - PROBLEM 2

1. **MOLECULAR FORMULA**

$C_{12}H_{22}O_3$

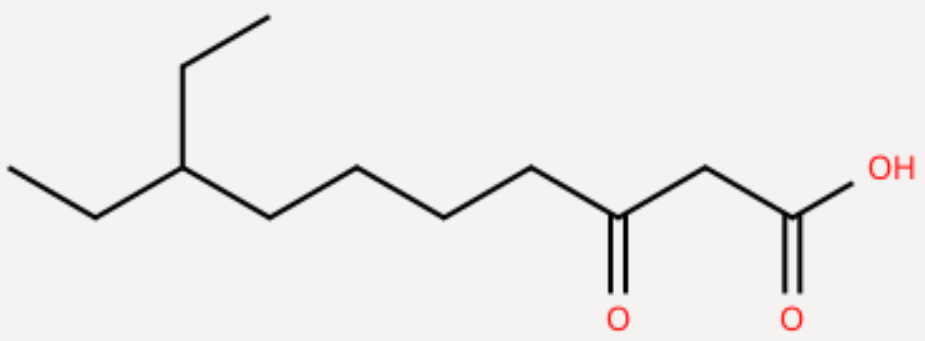
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$CH_3CH_2CH(C_2H_5)(CH_2)_4COCH_2COOH$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

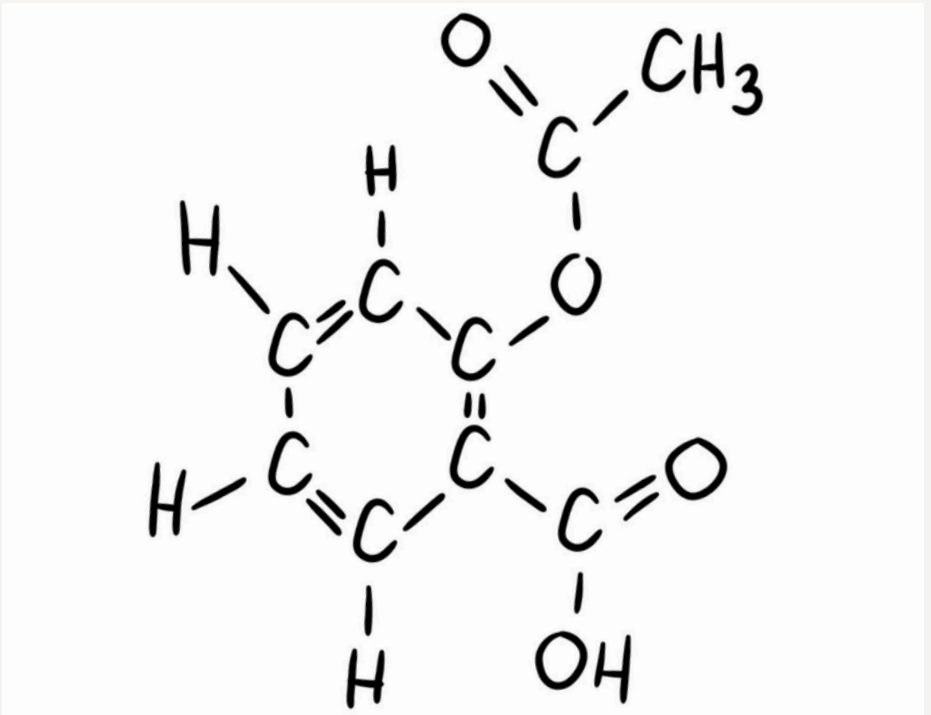
ACETYLSALICYLIC ACID

CARD 13 - PROBLEM 3

1. **MOLECULAR FORMULA**

$C_9H_8O_4$

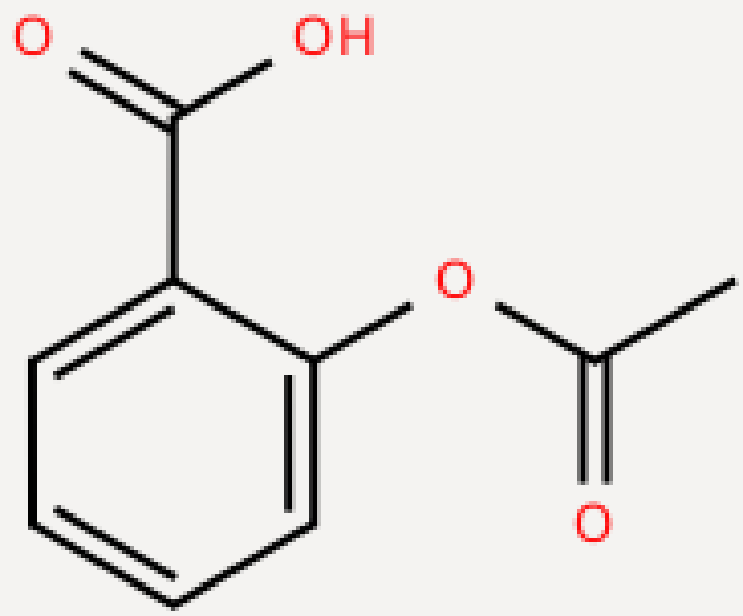
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$CH_3COOC_6H_4COOH$

4. **BOND-LINE FORMULA**



VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

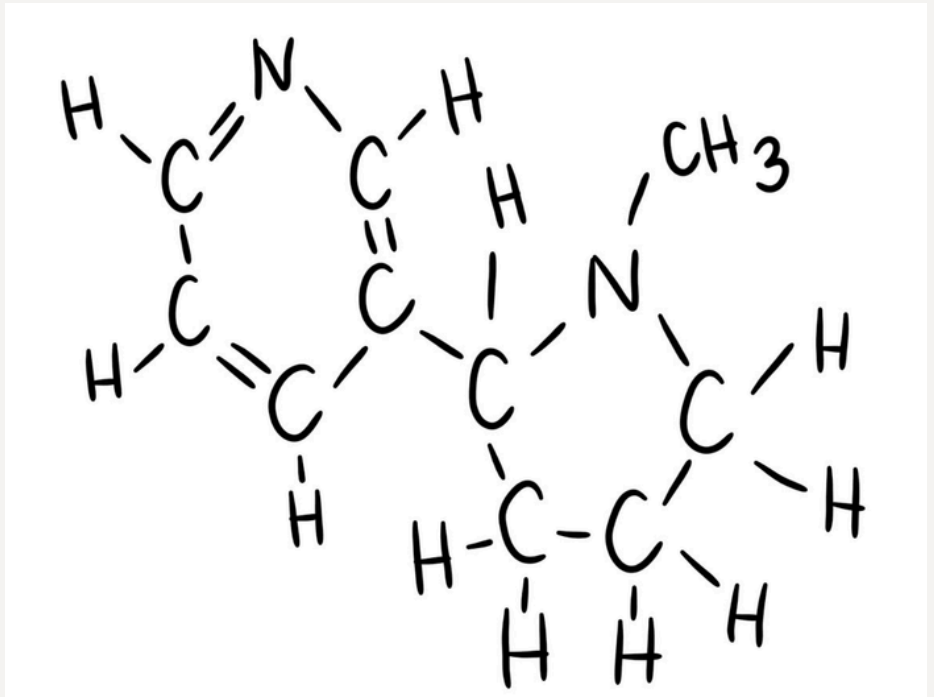
NICOTINE

CARD 14 - PROBLEM 4

1. **MOLECULAR FORMULA**

$C_{10}H_{14}N_2$

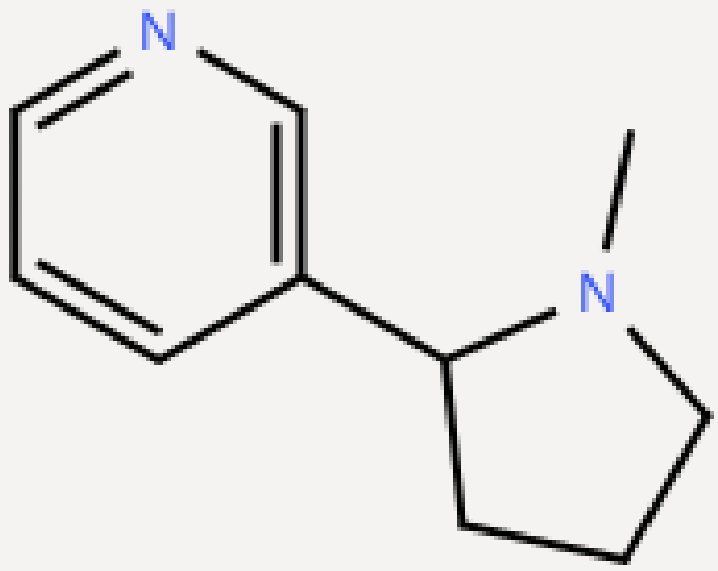
2. **LEWIS STRUCTURE**



3. **CONDENSED STRUCTURAL FORMULA**

$C_5H_4N–C_4H_7NCH_3$

4. **BOND-LINE FORMULA**



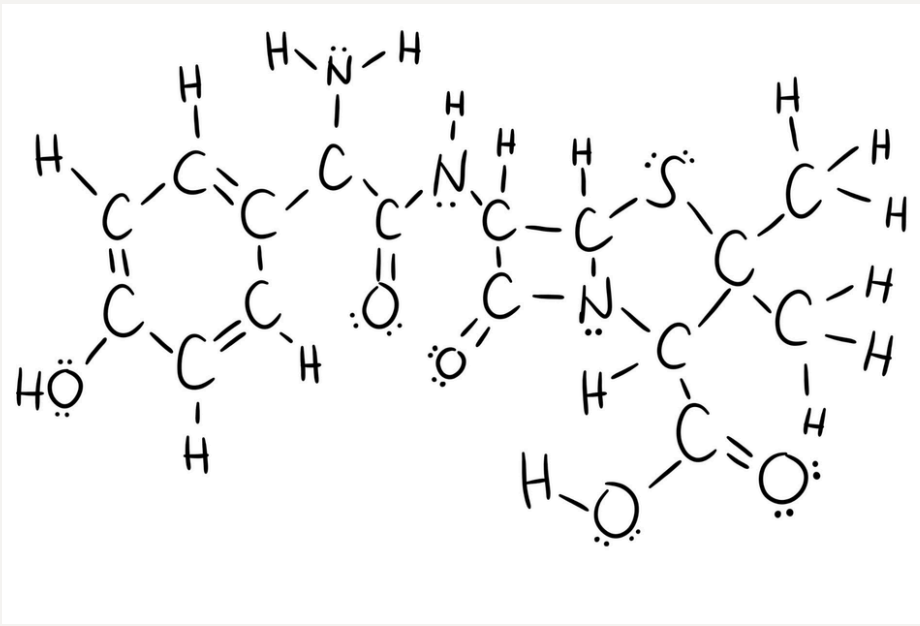
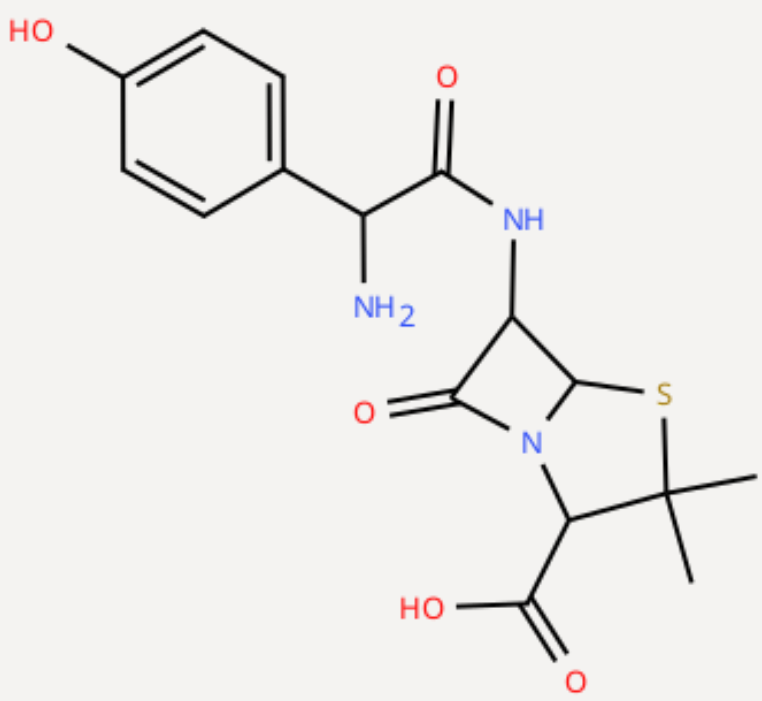
VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED

AMOXICILLIN

CARD 15 - PROBLEM 5

1. **MOLECULAR FORMULA**	2. **LEWIS STRUCTURE**
$C_{16}H_{19}N_3O_5S$	
3. **CONDENSED STRUCTURAL FORMULA**	4. **BOND-LINE FORMULA**
$(HOC_6H_4)CH(NH_2)CONH \cdot [C_4H_3NOS] \cdot C(CH_3)_2COOH$	

VERIFICATION STRIP

☐ ATOM COUNT VERIFIED ☐ VALENCE CHECKED

☐ BOND ODER CHECKED ☐ SAME CONNECTIVITY CONFIRMED